YUDI KARMA

SENIOR MOBILE & SECURITY SDK ENGINEER

Email: yudikarmateknik@gmail.com
Location: South Jakarta, Indonesia
LinkedIn: linkedin.com/in/yudikarma
GitHub: github.com/yudikarma | **Portfolio:** yudikarma.vercel.app

PROFESSIONAL SUMMARY

Senior Mobile Developer and Security SDK Specialist with over 8 years of software engineering experience architecting native Android applications, mobile banking security hardening, and real-time VoIP communications. Extensive enterprise experience consulting for national financial institutions including Bank Central Asia (BCA), Bank Negara Indonesia (BNI), Bank Tabungan Negara (BTN), CIMB Niaga, Panin Bank, and Bank BJB. Expert in Kotlin, Java, Jetpack Compose, MVVM/Clean Architecture, OneSpan Security, DexGuard RASP shielding, mTLS pin enforcement, Avaya WebRTC, and Tetherfi SDK integrations.

TECHNICAL COMPETENCIES

Mobile Engineering:	Kotlin, Java, Jetpack Compose, Android SDK, MVVM, Clean Architecture, Coroutines, Flow, Dagger Hilt, Room DB, WorkManager, Camera 1 & 2 APIs, EXIF Metadata
Security & Hardening:	OneSpan Mobile Security Suite, DexGuard (RASP & Obfuscation), Soft Token Authentication, Cryptographic Signing, mTLS, Root & Tamper Detection
VoIP & Omnichannel:	Avaya Elite WebRTC & Oceana SDK, Tetherfi Customer Engagement SDK, SIP Signaling, WebSockets, Audio/Video Stream Tuning
APM & Utilities:	Dynatrace APM (Real User Monitoring & Telemetry), Flutter, Dart, React Native, RESTful APIs, SQLite, Git, Jira, Automated Testing

PROFESSIONAL EXPERIENCE

Senior Android Developer & Enterprise Consultant
Jun 2021 — Present

PT. Phintraco Technology — Enterprise Financial & Security Consulting

- **KeyBCA Soft Token (Client: Bank Central Asia):** Developed and maintained KeyBCA mobile soft token client featuring cryptographic authentication algorithms and mTLS API integration for enterprise transaction validation.
- **OneSpan Security Suite (Clients: Panin Bank & Bank BJB):** Consulted client development teams on BJB Digi and Panin Mobile for multi-factor authentication, biometric binding, dynamic device fingerprinting, and application shielding.
- **Avaya Elite WebRTC SDK (Clients: Bank Central Asia & Bank Negara Indonesia):** Engineered reference sample applications in Kotlin using Jetpack Compose UI for Halo BCA and Wondr BNI demonstrating WebRTC media stream handling and SIP signaling.
- **Tetherfi Customer Engagement SDK (Clients: Bank BTN & CIMB Niaga):** Integrated Tetherfi SDK into bale by BTN and OCTO Mobile (CIMB Niaga), delivering secure Jetpack Compose UI sample components for native in-app customer chat and agent routing.
- **Security Hardening:** Enforced enterprise-level application shielding utilizing DexGuard dynamic obfuscation, root-detection algorithms, and mutual TLS (mTLS) certificate pinning.
- **Dynatrace APM & Telemetry:** Configured Dynatrace Real User Monitoring (RUM), performance telemetry dashboards, and real-time crash alerting for banking mobile clients.

Android Developer & Core Mobile Engineer
Nov 2019 — Jun 2021

PT. Derap Kreasi Persada — Full-Time Mobile Engineering

- **Santriqu App (Client: Educational Institutions):** Architected school fintech application using MVVM and Clean Architecture principles, integrating secure local payment gateways for PPOB and tuition processing.
- **Yipy Operations (Client: Centratama Group Staff):** Developed Flutter & Dart cross-platform application for facility management staff, handling real-time ticket updates over WebSockets and Google Maps logistics.
- **Warung Babe (Client: Giropos / PT Pos Indonesia Merchant Network):** Built native Android P2P gold trading client with Giropos API billing integration and reactive state updates for live market price feeds.

Native Android Developer
Nov 2018 — Nov 2019

PT. 107, BSD — Full-Time Mobile Developer

- **Snow Remove Camera Utility (Client: Industrial Snow Removal Contractor, Japan):** Developed specialized industrial camera application for snow removal workers in Japan using low-level Camera 1 and Camera 2 APIs, reverse-engineering raw open-source code to handle device-specific kernel crashes.
- **Zonapets Social Platform (Client: Hyrax Pet Network):** Engineered media sharing social app for pet owners, implementing image compression pipelines and nested RecyclerView feed caching.
- **EXIF Authentication:** Integrated custom EXIF location metadata injection into captured images to verify authentic field operational data.

Software Quality Assurance Intern (E-Procurement) — 2018 (6 Months)

PT. Mitra Mandiri Informatika — E-Procurement Systems

- **Promise Platform (Client: Enterprise E-Procurement):** Authored comprehensive manual test plans and automated regression scripts for enterprise e-procurement management systems on Jira.

DETAILED PROJECTS & TECHNICAL SPECIFICATIONS

KeyBCA (Individu & Bisnis) — Mobile Banking Soft Token

Client App: KeyBCA | Client: Bank Central Asia (BCA) | Employer: PT. Phintraco Technology

Tech Stack: Kotlin, Android SDK, OneSpan Security Suite, DexGuard RASP, mTLS, Cryptography

Developed and maintained banking soft token application to authenticate financial transactions. Implemented cryptographic token generation algorithms complying with strict banking regulatory standards. Integrated with internal BCA microservices over mTLS and enforced dynamic code obfuscation using DexGuard to mitigate reverse engineering and tampering risks.

Halo BCA & Wondr BNI — Avaya WebRTC Integration

Client Apps: Halo BCA (BCA) & Wondr BNI (BNI) | Employer: PT. Phintraco Technology

Tech Stack: Kotlin, Jetpack Compose, Avaya Elite WebRTC SDK, Oceana, SIP Signaling

Technical consultant for Avaya Elite WebRTC SDK integrations. Engineered customized reference sample applications in Kotlin using Jetpack Compose UI demonstrating WebRTC audio/video calling and SIP signaling for Halo BCA and Wondr BNI mobile banking applications.

bale by BTN & OCTO Mobile — Tetherfi Customer Engagement Integration

Client Apps: bale by BTN (Bank BTN) & OCTO Mobile (CIMB Niaga) | Employer: PT. Phintraco Technology

Tech Stack: Kotlin, Jetpack Compose, Tetherfi SDK, WebSockets, In-App Customer Engagement

Integrated Tetherfi customer engagement SDKs inside mobile banking platforms for Bank BTN (bale by BTN) and CIMB Niaga (OCTO Mobile). Delivered modular Kotlin and Jetpack Compose UI components for interactive real-time messaging, notification queues, and custom brand styling.

BJB Digi & Panin Mobile — OneSpan Security Integration

Client Apps: BJB Digi (Bank BJB) & Panin Mobile (Panin Bank) | Employer: PT. Phintraco Technology

Tech Stack: Kotlin, Jetpack Compose, OneSpan Security Suite, Biometric Binding, RASP

Led security implementation consulting for ONE SPAN Mobile Security Suite. Created Kotlin and Jetpack Compose sample applications to guide banking clients on integrating multi-factor authentication, biometric binding, and device fingerprinting to comply with financial regulatory requirements.

PROFESSIONAL CERTIFICATIONS

- **Google Cloud Certified Associate Cloud Engineer** — Google Cloud (May 2025 — May 2028)
- **Google Cloud Certified Cloud Digital Leader** — Google Cloud (Mar 2025 — Mar 2028)
- **Dynatrace Certified Associate** — Dynatrace (Mar 2026 — Mar 2028)
- **Dynatrace Certified Essentials** — Dynatrace (Mar 2026 — Mar 2028)
- **Avaya Infinity™ Design and Workflows** — Avaya (Sep 2025 — Lifetime)
- **Google Developer Kejar (Android Developer)** — Google x Dicoding (2019)
- **Technical Support Certification** — Balai Pelatihan Kerja (2018)

EDUCATION

Politeknik Aceh — Bachelor of Applied Science in Informatic Engineering (GPA: 3.56/4.00) — **2014 — 2018**